

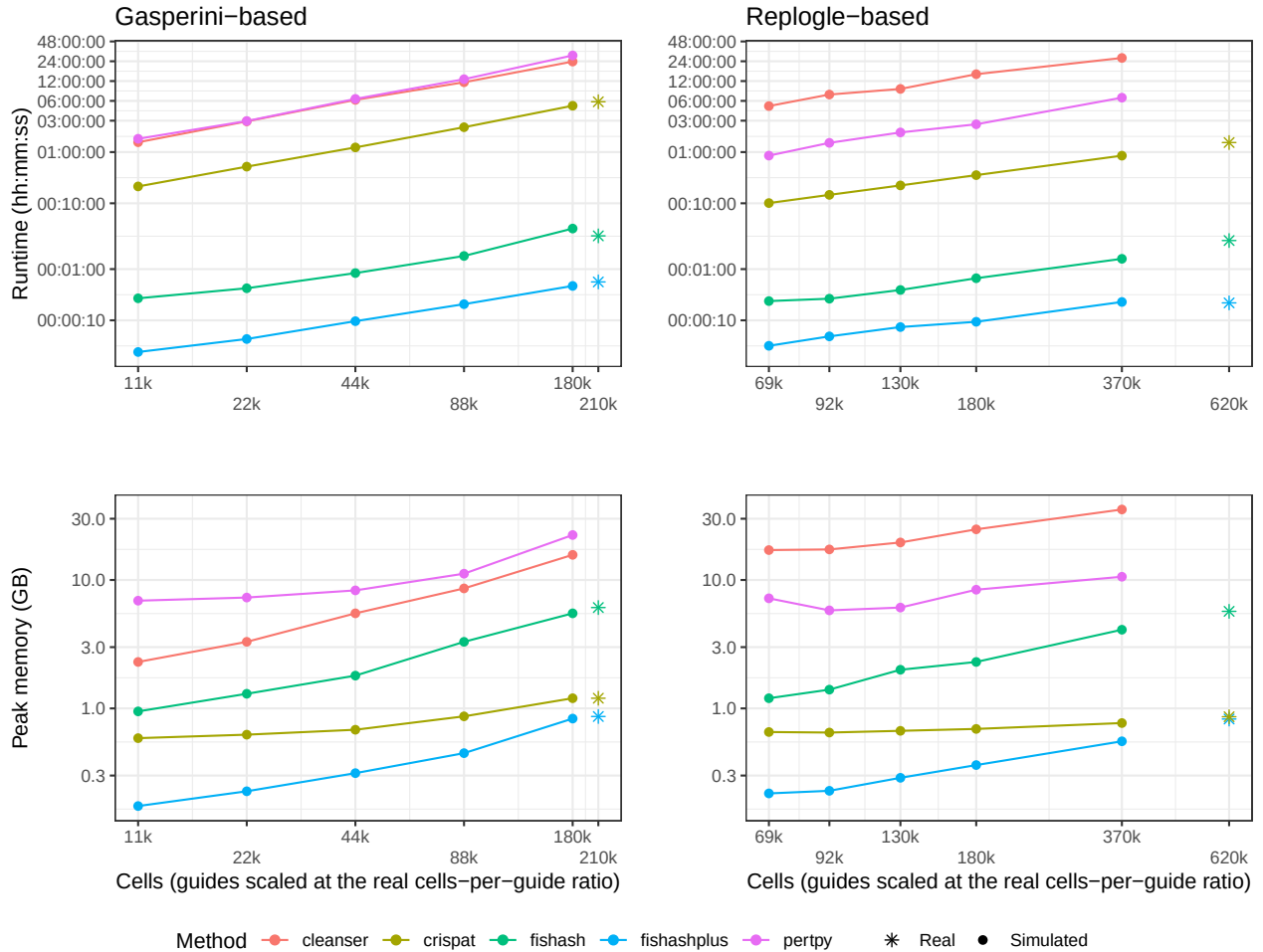
meeting-2026-10-08

2026-10-05

1. bigger computational results

main takeaways

- fishash+ is by far the fastest
- fishash isn't as great for memory
- fishash+ and crispat end up basically agreeing for memory. This will be included in memory investigation.
- our curves basically end at the real datasets, so that is good for our scaling datasets.



statistical benchmarking

Gene did an experiment a month ago, where he varied MOI and showed that fishash+ increasingly outperforms fishash.

In GuideBender, if we vary *only* the MOI parameter λ (which is the rate of the Poisson in the Hurdle-Poisson, not the actual expected MOI), we get other behavior changes. When λ increases in GuideBender, with all else (including `snr` and `count_per_cell`) held fixed, then there is a decrease in the typical UMI count for each infection. The expected per-cell signal total stays fixed because we are adding up more smaller UMI counts. The per-cell SNR [the ratio of expected per-cell signal to expected per-cell noise] is unchanged, but the per-guide SNR decreases as the problem becomes less separated in each guide.

The math of the model shows this too, where `d_mu_cell_drop_diff` (the expected difference between a signal cell-level scaling factor d_c^{cell} and endogenous error magnitude d_c^{drop}) is equal to $-\log(\text{avg_infections})$ up to a constant in λ . This means that doubling `avg_infections` makes `d_mu_cell_drop_diff` smaller by $\log 2$. Signal and endo are less separated.

This also increases the amount of “co-occurrences”, where a cell expressing many guides has a very high expected count in fishash, and so recall suffers.

So in GuideBender, I want to vary MOI separate from the hardness of each problem. For my statistical benchmarking, I want a univariate measure of problem hardness. With all else fixed, Claude proposed `ambient_per_cell`, the expected total ambient (endogenous) noise for each cell, and I like this so far. I want the mixture models to overlap more, and increasing the ambient magnitude is a reasonable way to do so.

These parameters interact, as described above, so we need to take that into account when determining their relationship. In particular, `snr` and `count_per_cell` will change as either MOI or `ambient_per_cell` change.

I am running this right now.